```
import os
from google.colab import drive
from google.colab import userdata

# 1. Mount Google Drive for persistent storage
drive.mount('/content/drive')

# 2. Define a persistent workspace in your Drive
WORKSPACE_DIR = '/content/drive/MyDrive/Colab_Workspace'
os.makedirs(WORKSPACE_DIR, exist_ok=True)
os.chdir(WORKSPACE_DIR)

# 3. Retrieve the secured GitHub token using your specific name
try:
    GITHUB_TOKEN = userdata.get('PAT')
    print("✅ GitHub Secret retrieved successfully!")
except userdata.SecretNotFoundError:
    print("❌ Error: Could not find 'PAT' in Secrets. Check spelling and capitalization.")
    GITHUB_TOKEN = None

# Only proceed if the token was found to prevent NameErrors
if GITHUB_TOKEN:
    # 4. Clone or update the repository
    REPO_NAME = 'AI-Native-News-Experience'
    REPO_URL = f"https://{GITHUB_TOKEN}@github.com/tiwaritanay01/{REPO_NAME}.git"

    if not os.path.exists(REPO_NAME):
        print(f"Cloning {REPO_NAME} directly into your Google Drive...")
        !git clone {REPO_URL}
    else:
        print(f"Directory {REPO_NAME} already exists in Drive. Pulling latest changes...")
        os.chdir(REPO_NAME)
        !git pull origin main  # Change 'main' if your default branch is different
        os.chdir('..')

    # 5. Install dependencies
    os.chdir(REPO_NAME)
    if os.path.exists("requirements.txt"):
        !pip install -r requirements.txt

    !pip install fastapi uvicorn pyngrok
    print("✅ Setup complete! Your code is safely backed up in Google Drive.")
else:
    print("⚠️ Setup paused. Please ensure your GitHub token is saved in Secrets exactly as '
```

```
Mounted at /content/drive
✅ GitHub Secret retrieved successfully!
Directory AI-Native-News-Experience already exists in Drive. Pulling latest changes...
fatal: couldn't find remote ref main
Requirement already satisfied: fastapi in /usr/local/lib/python3.12/dist-packages (0.135.1)
Requirement already satisfied: uvicorn in /usr/local/lib/python3.12/dist-packages (0.42.0)
Requirement already satisfied: pyngrok in /usr/local/lib/python3.12/dist-packages (7.5.1)
Requirement already satisfied: starlette>=0.46.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pydantic>=2.7.0 in /usr/local/lib/python3.12/dist-packages (f
Requirement already satisfied: typing-extensions>=4.8.0 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: click>=7.0 in /usr/local/lib/python3.12/dist-packages (from u
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.12/dist-packages (from uvi
Requirement already satisfied: PyYAML>=5.1 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: anyio<5,>=3.6.2 in /usr/local/lib/python3.12/dist-packages (f
```

```
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from an
✅ Setup complete! Your code is safely backed up in Google Drive.
```

```
import os
import shutil
import subprocess
import time
import requests
from pyngrok import ngrok
from google.colab import userdata

# 1. Reset Environment
%cd /content
if os.path.exists('AI-Native-News-Experience'):
    shutil.rmtree('AI-Native-News-Experience')

# 2. Pull Fresh Code (The one I just fixed locally)
GITHUB_TOKEN = userdata.get('PAT')
GITHUB_REPO = "github.com/tiwaritanay01/AI-Native-News-Experience.git"
!git clone https://{GITHUB_TOKEN}@{GITHUB_REPO}

# 3. Entry Point
os.chdir('/content/AI-Native-News-Experience/News_Backend')

# 4. Clean Stale Processes
!pkill -9 uvicorn
!pkill -9 ngrok

# 5. Start Server
PORT = 8000
print(f"Starting FastAPI on port {PORT}...")
with open("uvicorn.log", "w") as log_file:
    process = subprocess.Popen(
        ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", str(PORT)],
        stdout=log_file, stderr=log_file, text=True
    )

# 6. Final Health Check & Tunneling
print("Waiting for AI Engines (FinBERT, MiniLM) to initialize...")
for i in range(40): # Increased timeout for initial model download
    try:
        r = requests.get(f"http://127.0.0.1:{PORT}/")
        # 404 is okay (it means the server is up but root is empty)
        if r.status_code in [200, 404]:
            print("\n✅ AI BACKEND IS LIVE!")
            ngrok.set_auth_token(userdata.get('Ngrok'))
            public_url = ngrok.connect(PORT, "http")
            print("-" * 50)
            print(f"🚀 PUBLIC URL: {public_url}")
            print("-" * 50)
            break
    except:
        print(".", end="")
        time.sleep(3)
```

```
/content
Cloning into 'AI-Native-News-Experience'...
remote: Enumerating objects: 189, done.
remote: Counting objects: 100% (189/189), done.
remote: Compressing objects: 100% (141/141), done.
remote: Total 189 (delta 61), reused 168 (delta 40), pack-reused 0 (from 0)
Receiving objects: 100% (189/189), 457.63 KiB | 14.76 MiB/s, done.
```

```
Resolving deltas: 100% (61/61), done.
Starting FastAPI on port 8000...
Waiting for AI Engines (FinBERT, MiniLM) to initialize...
..........
✅ AI BACKEND IS LIVE!
--------------------------------------------------
🚀 PUBLIC URL: NgrokTunnel: "https://hypernatural-phoebe-blowy.ngrok-free.dev" -> "http://lo
--------------------------------------------------
```

```
# See the final few lines of your server logs
!tail -n 20 uvicorn.log
```

```
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you
Stored 109 articles in vector DB
Vector database saved.
Loading weights: 100%|██████████| 103/103 [00:00<00:00, 3582.47it/s]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you
INFO:     Started server process [19260]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:     127.0.0.1:58990 - "GET / HTTP/1.1" 200 OK
INFO:     14.143.83.158:0 - "GET / HTTP/1.1" 200 OK
```

```
# See the live server logs as they happen
!tail -f uvicorn.log
```

```
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you
INFO:     Started server process [19260]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:     127.0.0.1:58990 - "GET / HTTP/1.1" 200 OK
INFO:     14.143.83.158:0 - "GET / HTTP/1.1" 200 OK
INFO:     14.143.83.158:0 - "GET / HTTP/1.1" 200 OK
^C
```

```
# Fix the function call on line 23 in Colab
with open('/content/AI-Native-News-Experience/News_Backend/main.py', 'r', encoding='utf-8')
    code = f.read()

# Replace the incorrect call with the correct one
fixed_code = code.replace('detect_story_of_day()', 'detect_story_of_the_day()')

with open('/content/AI-Native-News-Experience/News_Backend/main.py', 'w', encoding='utf-8')
    f.write(fixed_code)

print("✅ Function call fixed! You can now re-run the backend cell.")
```

```
✅ Function call fixed! You can now re-run the backend cell.
```

```
# Fix the typo in main.py directly in Colab
with open('/content/AI-Native-News-Experience/News_Backend/main.py', 'r', encoding='utf-8')
    code = f.read()

# Replace the incorrect name with the correct one
fixed_code = code.replace('detect_story_of_day', 'detect_story_of_the_day')

with open('/content/AI-Native-News-Experience/News_Backend/main.py', 'w', encoding='utf-8')
    f.write(fixed_code)

print("✅ Typo fixed in main.py! You can now re-run your Start Backend cell.")
```

```
✅ Typo fixed in main.py! You can now re-run your Start Backend cell.
```

```
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
# These imports must match your directory structure
try:
    from app.api.news_api import app as news_app
    app = news_app
except ImportError:
    # Fallback if imports are tricky during setup
    app = FastAPI()

# 1. Essential CORS for frontend and ngrok to work
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_methods=["*"],
    allow_headers=["*"],
)

# 2. Add a explicit Health Check route at the root
@app.get("/")
def read_root():
    return {"status": "online", "message": "AI Native News Backend is Active"}

# 3. Existing routes (ensure these don't conflict)
@app.get("/health")
def health():
    return {"status": "ok"}
```

```
/usr/local/lib/python3.12/dist-packages/hdbscan/robust_single_linkage_.py:175: SyntaxWarning
  $max \{ core_k(a), core_k(b), 1/\alpha d(a,b) \}$.
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://hugg
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or
  warnings.warn(
config.json: 100%  758/758 [00:00<00:00, 51.2kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the
pytorch_model.bin: 100%  438M/438M [00:07<00:00, 2.18GB/s]
model.safetensors: 15%  67.0M/438M [00:02<00:01, 247MB/s]
Loading weights: 100%  201/201 [00:00<00:00, 4128.35it/s]
BertForSequenceClassification LOAD REPORT from: ProsusAI/finbert
Key                          | Status     |  |
-----------------------------+------------+--+-
bert.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you
tokenizer_config.json: 100%  252/252 [00:00<00:00, 9.65kB/s]
vocab.txt: 232k/? [00:00<00:00, 2.55MB/s]
special_tokens_map.json: 100%  112/112 [00:00<00:00, 1.17kB/s]
```

```
# See which models are taking up space/CPU
!ps -aux | grep uvicorn
```

```
root        7670  8.0  0.0      0     0 ?        Z    09:01   0:26 [uvicorn] <defunct>
root        9071  0.0  0.0   7372  3440 ?        S    09:07   0:00 /bin/bash -c ps -aux | gr
root        9073  0.0  0.0   6480  2448 ?        S    09:07   0:00 grep uvicorn
```

```
import os
import shutil

# 1. Step back to the root folder (fixes the 'no such file or directory' shell error)
%cd /content

# 2. Cleanly wipe the folder if it partially exists
if os.path.exists('AI-Native-News-Experience'):
    shutil.rmtree('AI-Native-News-Experience')
    print("Cleaned existing directory.")

# 3. Pull the fresh code
from google.colab import userdata
GITHUB_TOKEN = userdata.get('PAT')
GITHUB_REPO = "github.com/tiwaritanay01/AI-Native-News-Experience.git"

print("Re-cloning from GitHub...")
!git clone https://{GITHUB_TOKEN}@{GITHUB_REPO}

# 4. Check the content
print("\nVerifying files in News_Backend/app/services/:")
!ls /content/AI-Native-News-Experience/News_Backend/app/services/
```

```
/content
Re-cloning from GitHub...
Cloning into 'AI-Native-News-Experience'...
remote: Enumerating objects: 173, done.
remote: Counting objects: 100% (173/173), done.
remote: Compressing objects: 100% (129/129), done.
remote: Total 173 (delta 50), reused 159 (delta 36), pack-reused 0 (from 0)
Receiving objects: 100% (173/173), 455.40 KiB | 12.65 MiB/s, done.
Resolving deltas: 100% (50/50), done.

Verifying files in News_Backend/app/services/:
briefing_service.py         sentiment_service.py
briefing_service_utf8.py    store_articles.py
debate_generator.py         story_cluster.py
debate_service.py           story_cluster_utf8.py
debug_db.py                 story_detector.py
deep_briefing.py            story_detector_utf8.py
embeddings.py               story_intelligence.py
entity_extractor.py         story_of_day.py
entity_extractor_utf8.py    story_timeline.py
impact_analysis.py          story_title.py
impact_analysis_utf8.py     story_title_utf8.py
impact_service.py           test_ai_intelligence.py
impact_simulator.py         test_ai_news_engine.py
llm_service.py              test_briefing.py
llm_service_utf8.py         test_cluster.py
news_briefing.py            test_news_intelligence.py
opinion_service.py          test_search.py
personalization.py          test_story_intelligence.py
question_answer_service.py  timeline_service.py
question_generator.py       vector_search.py
```

```
# See which models are taking up space/CPU
!ps -aux | grep uvicorn
```

```
root        7670  8.3  0.0      0     0 ?        Z    09:01   0:26 [uvicorn] <defunct>
root        9024  0.0  0.0   7372  3476 ?        S    09:07   0:00 /bin/bash -c ps -aux | gr
root        9026  0.0  0.0   6480  2492 ?        S    09:07   0:00 grep uvicorn
```

```
import os

# 1. Go to the exact folder where main.py lives
WORKSPACE_DIR = '/content/drive/MyDrive/Colab_Workspace/AI-Native-News-Experience/News_Back
os.chdir(WORKSPACE_DIR)

# 2. Run Uvicorn directly in the foreground
!uvicorn main:app --host 127.0.0.1 --port 8000
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to e
Loading weights: 100% 103/103 [00:00<00:00, 1034.08it/s, Materializing param=pooler.dense.w
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if yo
Stored 108 articles in vector DB
Vector database saved.
Loading weights: 100% 103/103 [00:00<00:00, 962.58it/s, Materializing param=pooler.dense.we
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |
```

```
Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if yo
INFO:     Started server process [24043]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Shutting down
INFO:     Finished server process [24043]
ERROR:    Traceback (most recent call last):
  File "/usr/lib/python3.12/asyncio/runners.py", line 195, in run
    return runner.run(main)
           ^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/asyncio/runners.py", line 118, in run
    return self._loop.run_until_complete(task)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "uvloop/loop.pyx", line 1512, in uvloop.loop.Loop.run_until_complete
  File "uvloop/loop.pyx", line 1505, in uvloop.loop.Loop.run_until_complete
  File "uvloop/loop.pyx", line 1379, in uvloop.loop.Loop.run_forever
  File "uvloop/loop.pyx", line 557, in uvloop.loop.Loop._run
  File "uvloop/loop.pyx", line 476, in uvloop.loop.Loop._on_idle
  File "uvloop/cbhandles.pyx", line 83, in uvloop.loop.Handle._run
  File "uvloop/cbhandles.pyx", line 63, in uvloop.loop.Handle._run
  File "/usr/local/lib/python3.12/dist-packages/uvicorn/server.py", line 78, in serve
    with self.capture_signals():
         ^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/contextlib.py", line 144, in __exit__
    next(self.gen)
  File "/usr/local/lib/python3.12/dist-packages/uvicorn/server.py", line 339, in capture_si
    signal.raise_signal(captured_signal)
  File "/usr/lib/python3.12/asyncio/runners.py", line 157, in _on_sigint
    raise KeyboardInterrupt()
KeyboardInterrupt

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/starlette/routing.py", line 701, in lifespa
    await receive()
  File "/usr/local/lib/python3.12/dist-packages/uvicorn/lifespan/on.py", line 137, in recei
    return await self.receive_queue.get()
```

```
import os

# Path to your main.py file
file_path = '/content/drive/MyDrive/Colab_Workspace/AI-Native-News-Experience/News_Backend/

if os.path.exists(file_path):
    # Read the contents of main.py
    with open(file_path, 'r', encoding='utf-8') as file:
        file_data = file.read()

    # Replace the incorrect name with the correct one everywhere in the file
    new_data = file_data.replace('detect_story_of_day', 'detect_story_of_the_day')

    # Save the file back
    with open(file_path, 'w', encoding='utf-8') as file:
        file.write(new_data)

    print("✅ Successfully updated 'detect_story_of_day' to 'detect_story_of_the_day' in ma
else:
    print("❌ Could not find main.py.")
```

```
✅ Successfully updated 'detect_story_of_day' to 'detect_story_of_the_day' in main.py!
```

```
import os
```

```
# 1. Make sure we are in your project folder
WORKSPACE_DIR = '/content/drive/MyDrive/Colab_Workspace/AI-Native-News-Experience'
os.chdir(WORKSPACE_DIR)

# 2. Run the server directly so we can see the crash logs
!uvicorn main:app --host 0.0.0.0 --port 8000
```

```
ERROR:    Error loading ASGI app. Could not import module "main".
```

```
import os

def fix_encoding(directory):
    for root, dirs, files in os.walk(directory):
        for file in files:
            if file.endswith('.py') or file == 'requirements.txt':
                file_path = os.path.join(root, file)
                try:
                    # Try reading as UTF-16
                    with open(file_path, 'rb') as f:
                        content = f.read()

                    # Detect if it's UTF-16 by checking for null bytes or BOM
                    if b'\x00' in content:
                        print(f"Converting {file_path} from UTF-16 to UTF-8...")
                        decoded_content = content.decode('utf-16')
                        with open(file_path, 'w', encoding='utf-8') as f:
                            f.write(decoded_content)
                except Exception as e:
                    # If it's already UTF-8 or something else, skip it
                    continue

# Run the fix on your cloned repo
fix_encoding('/content/AI-Native-News-Experience/News_Backend')
print("✅ Encoding fixed. You can now re-run the backend cell.")
```

```
Converting /content/AI-Native-News-Experience/News_Backend/requirements.txt from UTF-16 to U
Converting /content/AI-Native-News-Experience/News_Backend/main.py from UTF-16 to UTF-8...
✅ Encoding fixed. You can now re-run the backend cell.
```

```
!pip install chromadb
```

```
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 231.6/231.6 kB 20.4 MB/s eta 0:00:00
Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_1
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 71.6/71.6 kB 5.7 MB/s eta 0:00:00
Downloading pypika-0.51.1-py2.py3-none-any.whl (60 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 60.6/60.6 kB 5.4 MB/s eta 0:00:00
Downloading durationpy-0.10-py3-none-any.whl (3.9 kB)
Downloading pyproject_hooks-1.2.0-py3-none-any.whl (10 kB)
Installing collected packages: pypika, durationpy, pyproject_hooks, pybase64, opentelemetry
  Attempting uninstall: opentelemetry-proto
    Found existing installation: opentelemetry-proto 1.38.0
    Uninstalling opentelemetry-proto-1.38.0:
      Successfully uninstalled opentelemetry-proto-1.38.0
  Attempting uninstall: opentelemetry-exporter-otlp-proto-common
    Found existing installation: opentelemetry-exporter-otlp-proto-common 1.38.0
    Uninstalling opentelemetry-exporter-otlp-proto-common-1.38.0:
      Successfully uninstalled opentelemetry-exporter-otlp-proto-common-1.38.0
  Attempting uninstall: opentelemetry-api
    Found existing installation: opentelemetry-api 1.38.0
    Uninstalling opentelemetry-api-1.38.0:
      Successfully uninstalled opentelemetry-api-1.38.0
  Attempting uninstall: opentelemetry-semantic-conventions
    Found existing installation: opentelemetry-semantic-conventions 0.59b0
    Uninstalling opentelemetry-semantic-conventions-0.59b0:
      Successfully uninstalled opentelemetry-semantic-conventions-0.59b0
  Attempting uninstall: opentelemetry-sdk
    Found existing installation: opentelemetry-sdk 1.38.0
    Uninstalling opentelemetry-sdk-1.38.0:
      Successfully uninstalled opentelemetry-sdk-1.38.0
ERROR: pip's dependency resolver does not currently take into account all the packages that
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-exporter-otlp-proto-co
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-proto==1.38.0, but you
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-sdk~=1.38.0, but you h
opentelemetry-exporter-gcp-logging 1.11.0a0 requires opentelemetry-sdk<1.39.0,>=1.35.0, but
google-adk 1.27.1 requires opentelemetry-api<1.39.0,>=1.36.0, but you have opentelemetry-ap
google-adk 1.27.1 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you have opentelemetry-sc
Successfully installed bcrypt-5.0.0 build-1.4.0 chromadb-1.5.5 durationpy-0.10 kubernetes-3
```